

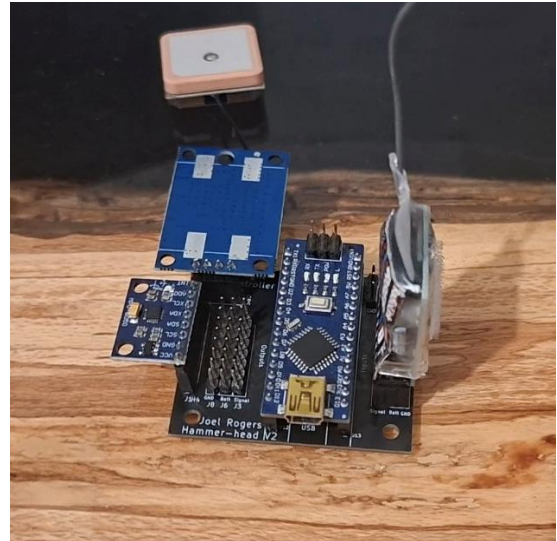
Joel Rogers

Mechatronics Engineering Portfolio

UAV V2



Fabricated an ultra-strong **fibre glass** core section (light grey) for strength, small wall thickness and weight.



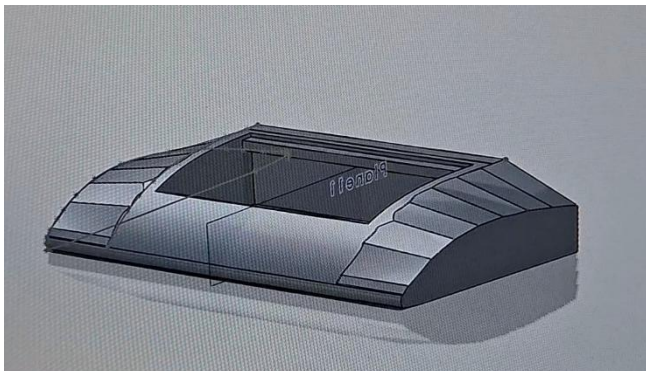
Designed a **flight controller (KiCAD)** with integrated GPS, RF receiver and IMU module for autonomous flight and fly by wire pilot assistance.



Hand launch of the UAV V2.



A display of the **fly by wire assistance** where the elevons are attempting to relevel the plane.



Designed, modelled (**SolidWorks**), manufactured and tested a flying wing UAV platform.



Design changes through the learning of the prototype (**design through iteration**).

UAV Prototype



Designed, modelled (**SolidWorks**), manufactured and tested a flying wing UAV platform.



Integrated long range **RF** communication for **command and control**.



Designing through failure and partial destruction (**Design through iteration**).

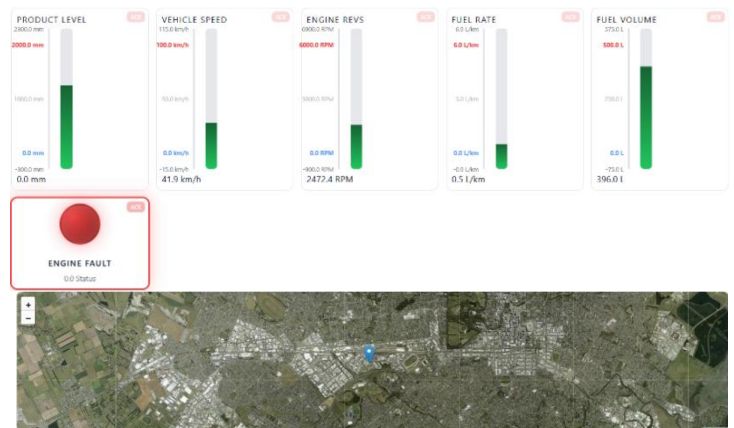


Achieved flight after six failed attempts and subsequent redesigns.

Sentinel Wireless IoT Module

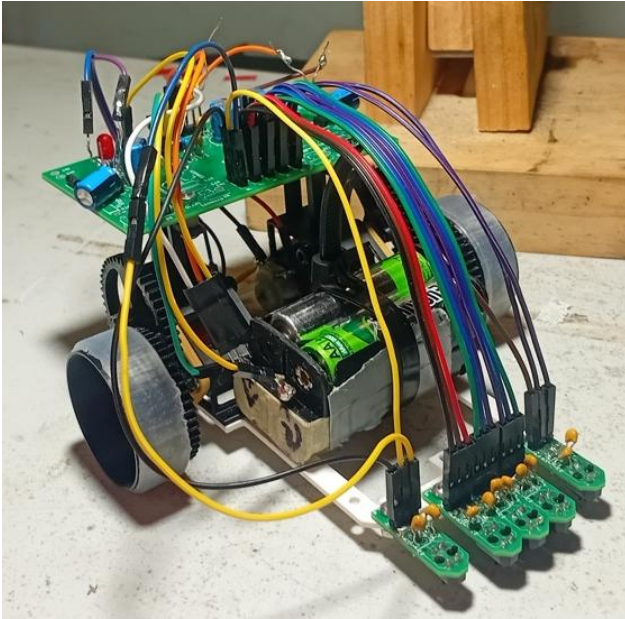


Multiprotocol wireless communicating IoT ESP32 gateway, communicating through **WiFi**, **4G**, **Ethernet** and obtaining **GPS telemetry**.

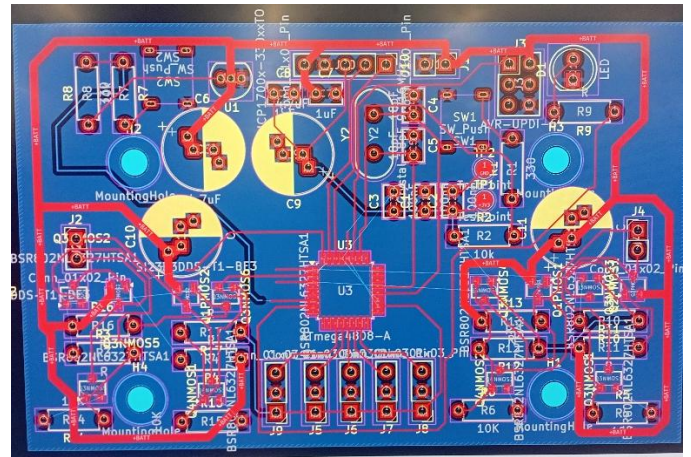


Also included the construction of a **server**, **database** and **web UI**.

Line Following Robot

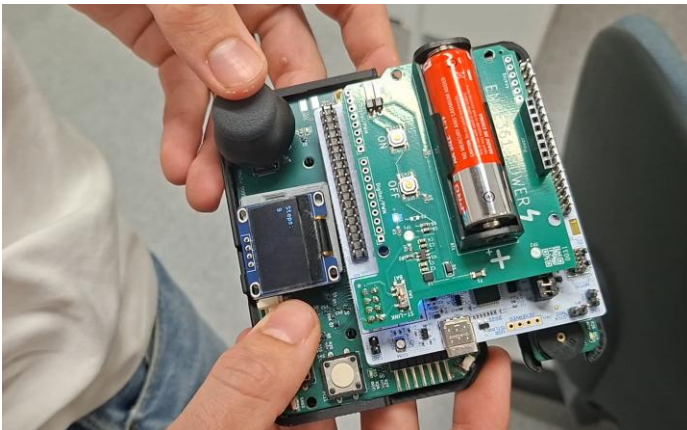


Broke the record for the annual University of Canterbury line following robot competition, with our fastest time being **5.81s**. Designed, 3D modelled (**Fusion 360/SolidWorks**), **3D printed** and tested numerous gearing systems, motor mounts and chassis.



Designed (**KiCAD**) and assembled the main **microcontroller PCB** and smaller sensor PCBs. Programmed the robot in **Arduino IDE** and integrated an ESP8266 for **wireless communication** and PID value changes over **WIFI**.

STM32 Step Counter



Programmed (**STM32CubeIDE**) an STM32 walking step counting system using **interrupt** driven software. Utilized **GIT** for version control.

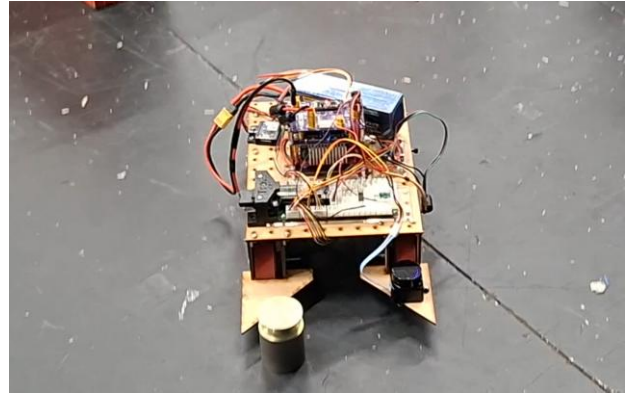


Steps were detected through reading an **IMU** and implementing threshold based counting.

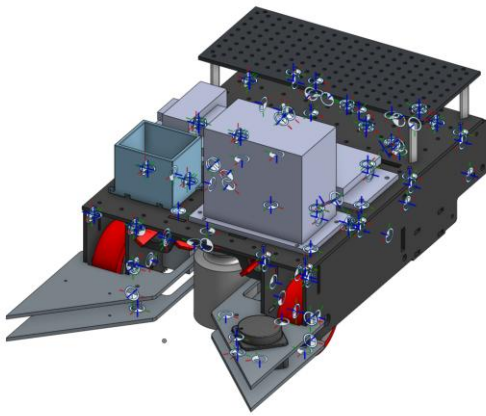
RoboCup



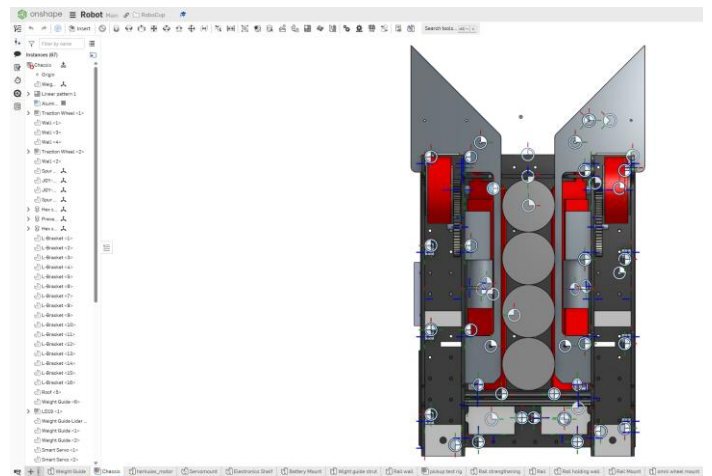
Used **first principles** engineering to design and **iterate** a weight collection system,



Developing a **MDF** chassis using **Lazer cutting** and **3d printing** to collect metal weights around an arena.



3d modelled using **Onshape**.



Using **Onshapes** version control for ease of **redesign** and **reverting changes**.

Joel Rogers

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PROFESSIONAL SUMMARY

Mechatronics Engineer at Swarm Intelligence and penultimate year **Mechatronics Engineering student** at the University of Canterbury, specialising in **GNSS/GPS**, **electronic hardware** and **wireless data communication (radio, WiFi and 4G)**. Experience designing and integrating **avionics** into fixed wing UAV platforms. Industry experience delivering **firmware programming (C++)**, **PCB design** and **system automation** in environments where reliability and performance are critical.

EXPERIENCE

Professional

Mechatronics Engineer, Swarm Intelligence

Christchurch, Canterbury, March 2026-Present

- Led development of an ESP32 based multi-protocol **wireless communication** system incorporating **GNSS tracking**, embedded **firmware**, **hardware integration** and backend **server development**.
- **PCB design (KiCAD)** for microcontroller and daughter boards in real world mobile systems.
- **Design and implementation** of automation and embedded control systems deployed in high-performance engineering and manufacturing environments (including projects with **Pratt & Whitney** and **Fulton Hogan**).
- Designing enclosures and mounting systems for electronics. (**SolidWorks**, **3D printing**)

Electronics/Robotics Engineering Intern, Swarm Intelligence

Christchurch, Canterbury, October 2025-March 2026

- **Soldered** over 50 PCBs, both surface mount and through hole.
- **Validated** all PCBs I soldered ensuring nominal operation before they were delivered to clients.
- On site **hardware installation and commissioning** teaching the safety requirements and physical debugging process when dealing with **high voltage**.

Tiler, Syrus Imperial Tiling Company LTD

Christchurch, Canterbury, December 2023- November 2025

- This experience allows me to combine a tradesman's **hands-on work ethic** with a strong analytical approach to bridge the gap between design and **real world implementation**.

Technical Projects

UAV (Personal)

- Designed a **flight controller (KiCAD)** with integrated **GPS** and **IMU** modules for **fly by wire** assistance and **autonomous** flight.
- Integrated an **ESC** motor controller and power regulating/distributing system.
- Designed through iteration, modelled (**SolidWorks**), manufactured and tested two flying wing UAV platforms, the prototype and the V2.
- Integrated long range **RF** communication for command and control.
- **Programmed** the flight controller to take inputs from the **RF**, **GPS** and **IMU** modules to control the plane accordingly.

Line Following Robot (University)

- **Broke the record** for the annual University of Canterbury line following robot competition, with our fastest time being **5.81s**.
- Designed (**KiCAD**) and assembled a **microcontroller PCB** and smaller photoelectric sensor PCBs.
- Programmed the robot in **Arduino IDE** and integrated an ESP8266 for **wireless communication** and PID value changes over **WiFi**.

STM32 Step Counter (University)

- Programmed (**STM32CubeIDE**) an STM32 walking step counting system using **interrupt** driven software.
- Steps were detected through reading an **IMU** and implementing threshold based counting.
- Utilized **GIT** for version control.

SKILLS

Programming:

- Python
- C, C++, C#, Arduino
- HTML
- JavaScript
- GIT

Embedded & Electronics:

- KiCAD
- Soldering and electronic assembly
- Sensor integration
- AutoCAD Electrical
- CX Programmer
- STM32CubeIDE
- Arduino IDE

Communication protocols and Networking:

- WiFi
- Ethernet
- SMS
- GNSS tracking
- MQTT
- Modbus
- Radio

Mechanical:

- SolidWorks
- 3D printing
- Fusion 360
- Lathe
- Mill

Soft Skills:

- **Leadership** from my time leading group projects at university, coaching hockey and taking charge of the bar at The Commodore hotel.
- **Relationship building** from interacting with customers at The Commodore Hotel and meeting new clients/coworkers at Swarm Intelligence.

Further Experience

- **Workplace First Aider** Certified (Valid until February 2028)

Professional:

- **Bartender**, The Commodore Hotel (November 2022 – February 2025)
- **Hockey Coach**, St Andrews College (April 2023-August 2025)
- **Representative Hockey Coach**, Canterbury Hockey (August 2025 – September 2025)
- **Representative Hockey Player**, Canterbury Hockey (Canterbury U18: April 2021-August 2023)

Technical Projects:

- **Designed and trained** multiple **AI lecture bots** to reattach me University content when revising for exams. (Personal)
- Analysed and tweaked in code a **machine learning model** for image edge detection. (University)
- Microcontroller LED **"Tic Tac Toe" game**. (University)
- PID Robot Cart (University)
- Aluminium Structure (University)
- **CNC** modular pick and place. (University)
- Arduino based **bike thief alarm**. (Personal)
- Arduino based **garage door sensor**. (Personal)
- Numerous Projects through my time at **Swarm Intelligence** (Swarm Intelligence)

EDUCATION

Bachelor of Engineering (Honours) in Mechatronics

Christchurch, University of Canterbury,

Expected graduation: 2027

Website: <https://j0llyr0ger.github.io/Joel-Rogers-Portfolio/>

References can be produced on request